Faecal retention: a common cause in functional bowel disorders, appendicitis and haemorrhoids--with medical and surgical therapy.

The present studies explored whether faecal retention in the colon is a causative factor in functional bowel disease, appendicitis, and haemorrhoids. Faecal retention was characterized by colon transit time (CTT) after radio-opaque marker ingestion and estimation of faecal loading on abdominal radiographs at 48 h and 96 h. Specific hypotheses were tested in patients (n = 251 plus 281) and in healthy random controls (n = 44). A questionnaire was completed for each patient, covering abdominal and anorectal symptoms and without a priori grouping. Patients with functional bowel disorders, predominantly women, had a significantly increased CTT and faecal load compared to controls. The CTT was significantly and positively correlated with segmental and total faecal loading. The faecal load was equal at 48 h and 96 h, mirroring the presence of permanent faecal reservoirs. In these first clinical studies to correlate bowel symptoms with CTT and colon faecal loading, abdominal bloating was significantly correlated with faecal loading in the right colon, total faecal load, and CTT. Abdominal pain was significantly and positively correlated to distal faecal loading and significantly associated with bloating. A new phenomenon with a high faecal load and a normal CTT was observed in a subset of patients (n = 90), proving faecal retention as hidden constipation. The CTT and faecal load were significantly higher in the right-side compared to the left and distal segments. Within the control group of healthy persons, the right-sided faecal load was significantly greater than the left and distal load. The CTT and faecal load significantly positively correlated with a palpable mass in the left iliac fossa and meteorism. Cluster analysis revealed that CTT and faecal load positively correlated with a symptom factor consisting of bloating, proctalgia and infrequent defecation of solid faeces. On the other hand, CTT and faecal load negatively correlated with a symptom factor comprising frequent easy defecations, repetitiveness, and incompleteness with solid or liquid faeces. The majority of patients with a heavy faecal load but normal CTT had repetitive daily defecation, mostly with ease and with altering faecal consistence. Flue-like episodes

co-existed in symptom factors with abdominal pain and meteorism, and these symptoms together with a palpable right iliac fossa mass and tenderness, and in other factors with seldom and difficult defecation, and with epigastric discomfort and halitosis. Patients with seldom and difficult defecation of solid faeces experienced abdominal pain significantly more often and presented a palpable mass in the right iliac fossa with tenderness and meteorism. The CTT was significantly prolonged and faecal load significantly increased. In patients with a normal CTT and increased faecal load, only patients with abdominal pain had a significant correlation between faecal loading and bloating. CTT and faecal load were shown for the first time to increase significantly with the number of colonic redundancies (colon length), which also resulted in significantly increased bloating and pain. Intervention with a bowel stimulation regimen combining a fibre-rich diet, fluid, physical activity, and a prokinetic drug was essential to proving that abdominal symptoms and defecation disorders are caused by faecal retention, with or without a prolonged CTT. The CTT was significantly reduced, as was faecal load. Bloating and pain were reduced significantly. The defecation became easy with solid faeces, towards one per day and with significant reductions in incompleteness and repetitiveness. Proctalgia and flue-like episodes were significantly reduced. The intervention significantly reduced the presence of a tender palpable mass in the right fossa and rectal constipation. In patients with a normal CTT but increased faecal load, the intervention did not significantly change the CTT or load, but bloating and pain were significantly reduced, just as defecation improved overall. The novel knowledge of faecal retention in the patients does not explain why faecal retention occurs. However, it may be inferred from the present results that a constipated or irritable bowel may belong to the same underlying disease dimension, where faecal retention is a common factor. Thus, measuring CTT and faecal load is suggested as a guide to a positive functional diagnosis of bowel disorders compared to the constellation of symptoms alone. Thirty-five patients underwent surgery after being refractory to the conservative treatment for constipation. They had a significantly prolonged CTT and heavy faecal loading, which was responsible for the aggravated abdominal and defaecatory symptoms. The operated patients presented with a redundant colon (dolichocolon) significantly more often. These patients also had an extremely high rate of previous appendectomy. Twenty-one patients underwent hemicolectomy, and 11 patients had a subtotal colectomy with an ileosigmoidal anastomosis; three patients received a stoma. However, some patients had to have the initial segmental colectomy converted to a final subtotal colectomy because of persisting symptoms. Six more subtotal colectomies have been performed and the leakage rate of all colectomies is then 4.9 % (one patient died). After a mean follow-up of 5 years, the vast majority of patients were without abdominal pain and bloating, having two to four defecations daily with control and their quality of life had increased considerably. A faecalith is often located in the appendix, the occlusion of which is responsible for many cases of acute appendicitis, which is infrequent in all except white populations. An effort to trace the origin of the faecalith to faecal retention in the colon was made in a case control study (56 patients and 44 random controls). The CTT was longer and faecal load greater in patients with appendicitis compared to controls, though the difference was not significant. Power calculations showed that more patients were needed to reach statistical significance for these parameters. The presence of a faecalith was most often associated with a gangrenous or perforated appendix. No significant differences were found between the CTT and faecal load of patients who had or did not have a faecalith. However, the right-sided faecal load was significantly higher than the left and distal load. Haemorrhoids are often a consequence of constipation and defaecatory disorders and were found in every second patient with functional bowel disorders. The present studies are the first Danish reports of a novel operation to cure this disease, stapled haemorrhoidopexy (n = 40 and 258 patients). The majority of patients had prolapsed haemorrhoids, and the durability of procedure was confirmed with a follow-up of up to 5 years, meaning a normal anus. The operation time was short, post-operative pain was low, and recovery was rapid. No incontinence was observed, and patient satisfaction was high and significantly correlated with the appearance of a normal anus without prolapse. The cumulative risk of re-operation was greatest in the first 2 years after the stapled haemorrhoidopexy. Patients with persisting haemorrhoidal prolapse had the procedure repeated with results as good as those obtained in the rest of the patients. It was shown in a statistical model that the preoperative severity of haemorrhoidal disease and the immediate postoperative result contributed significantly to predicting the outcome that is the durability of the operation. The most frequent post-operative complication was bleeding requiring surgical haemostasis. One serious complication occurred after an anastomotic leak from a highly placed anastomosis, resulting in retro rectal, retro- and intra-peritoneal, and mediastinal gas. The patient recovered after conservative treatment and without surgical intervention. The stapling technique now used has revolutionized the surgical treatment of prolapsing haemorrhoids. Finally, a common cause may be suspected for diseases constantly associated with one another. Epidemiological evidence has recognized that constipation, diverticulosis and IBS increase the risk of colon cancer (and adenomas), diseases exceedingly rare in communities exempt from appendicitis. Haemorrhoids are a colonic co-morbidity as well. Notably, the patients with a functional bowel disorder had a much higher rate of a previous appendectomy than the background population. In addition, the patients who had previously had an appendectomy had a significantly longer CTT compared to patients, who had not. The data points to the involvement of faecal retention in the origin of faecaliths and, thus, acute appendicitis. Faecal reservoirs were shown in the right and left colon segments in both patients and controls, which are the same areas bearing the highest incidences of adenomateous polyps and malignancies. Familial colorectal cancer occurred significantly more often in patients who had a higher faecal load than the controls. Four malignancies and 25 adenomas were identified. An increased faecal load in the colon with or without delayed transit will increase bacterial counts and create a chronic inflammation of the colonic mucosa, which is a risk factor for cancer onset. A functional bowel disorder is then likely to occur with gradually transition from a primary functional disease into specific organic diseases. A diet rich in fibre and regular physical activity have a therapeutic and preventive effect on colorectal diseases associated with faecal retention.